

R&D DOCUMENT ON THREE-TIER AZURE ARCHITECTURE DEPLOYMENT WITH VMs

PREPARED BY :

Richa Budhori

PURPOSE :

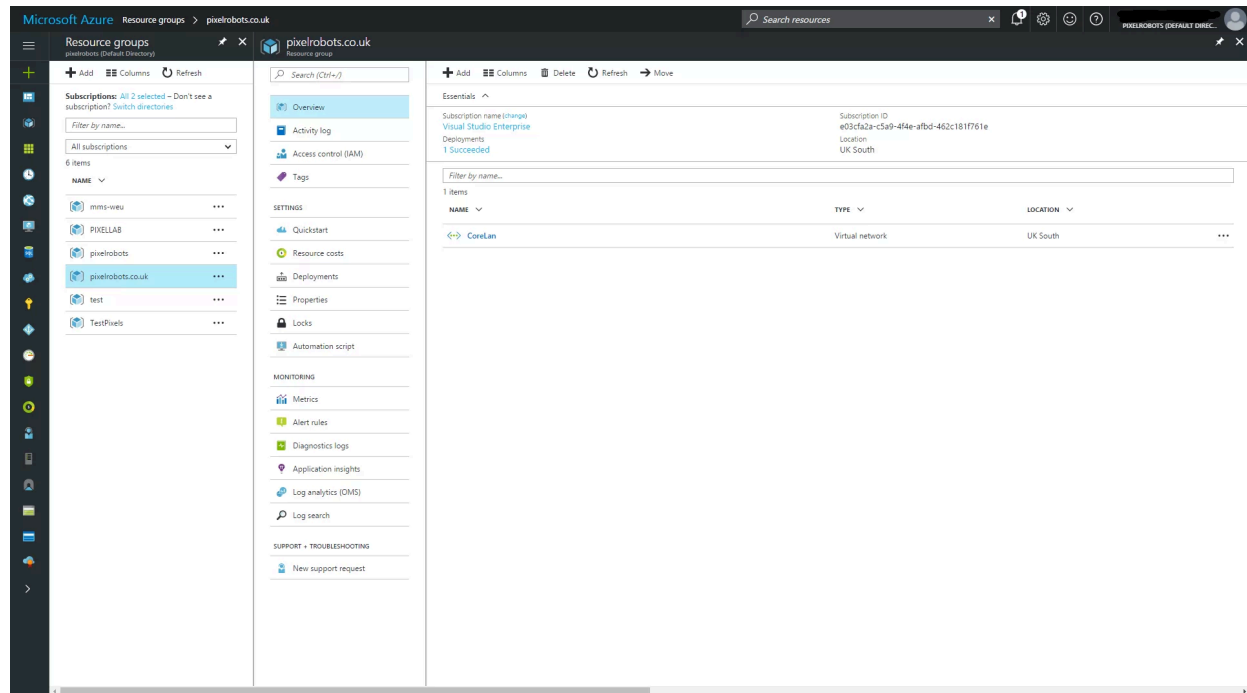
This document outlines the process of deploying a three-tier architecture (Web Tier, App Tier, and DB Tier) in Azure. It demonstrates network segmentation using subnets and controls access through Network Security Groups (NSGs).

ARCHITECTURAL DEPLOYMENT

Step 1: Sign In to Azure Portal

- Navigate to <https://portal.azure.com>.
- Log in with your Azure credentials.

Step 2: Create a Resource Group



Steps:

1. Search for “Resource groups” in the search bar → Click Create.
2. Enter the Resource Group name: ThreeTier-RG.
3. Choose your Region (e.g., East US).
4. Click Review + Create → Create.

Step 3: Create a Virtual Network (VNet)

Create virtual network ...



Basics **Security** IP addresses Tags Review + create

Enhance the security of your virtual network with these additional paid security services. [Learn more](#)

Virtual network encryption

Enable Virtual network encryption to encrypt traffic traveling within the virtual network. Virtual machines must have accelerated networking enabled. Traffic to public IP addresses is not encrypted. [Learn more](#)

Virtual network encryption ☐

Azure Bastion

Azure Bastion is a paid service that provides secure RDP/SSH connectivity to your virtual machines over TLS. When you connect via Azure Bastion, your virtual machines do not need a public IP address. [Learn more](#)

Enable Azure Bastion ☒

Azure Bastion host name

Azure Bastion public IP address *
[Create a public IP address](#)

Azure Firewall

Azure Firewall is a managed cloud-based network security service that protects your Azure Virtual Network resources. [Learn more](#)

Enable Azure Firewall ☐

Azure DDoS Network Protection

Azure DDoS Network Protection is a paid service that offers enhanced DDoS mitigation capabilities via adaptive tuning, attack notification, and telemetry to protect against the impacts of a DDoS attack for all protected resources within this virtual network. [Learn more](#)

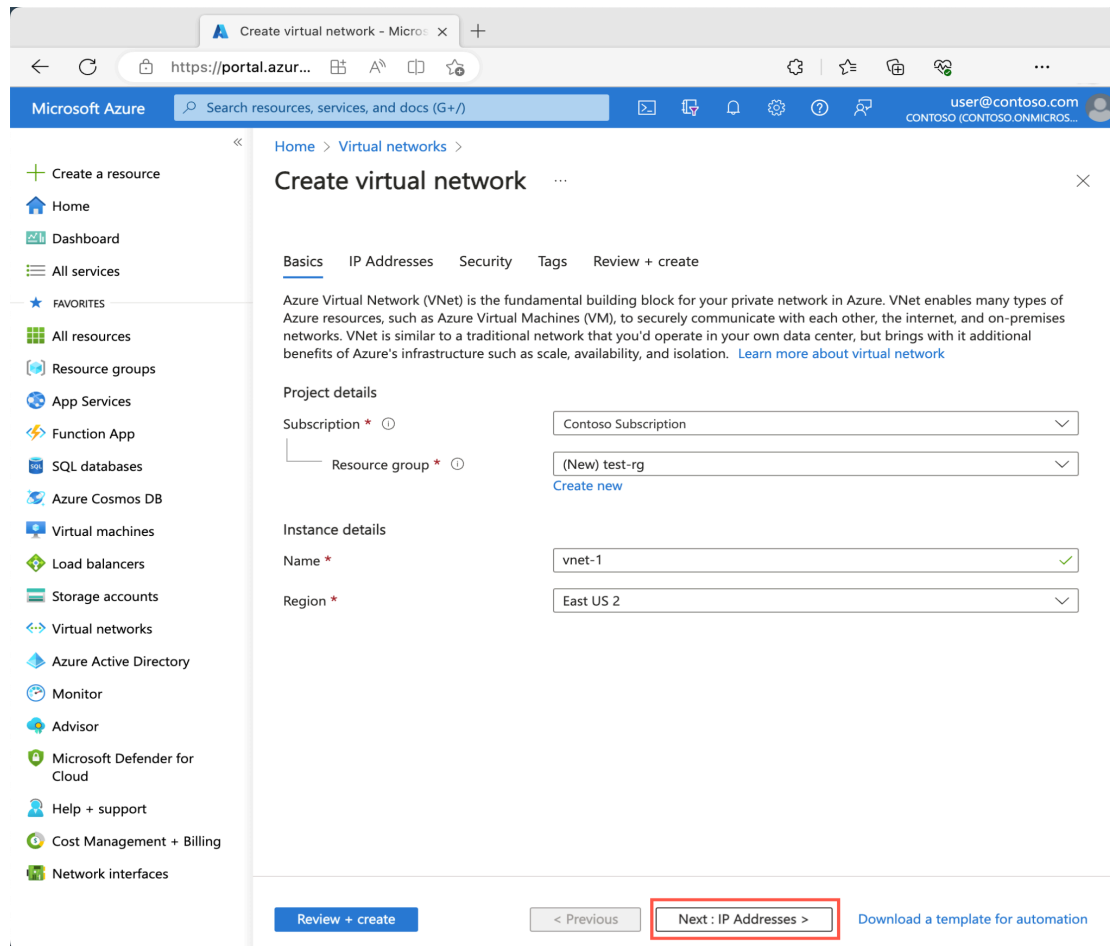
Enable Azure DDoS Network Protection ☐

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Steps:

1. Search for “Virtual networks” → Click Create.
2. Resource Group: Select ThreeTier-RG.
3. Name: ThreeTier-VNet.
4. Region: Same as the Resource Group.
5. IPv4 Address space: 10.0.0.0/16.
6. Click on Next: IP Addresses.

Step 4: Create Subnets

Home > Virtual networks >

Create virtual network ...

BasicsIP AddressesSecurityTagsReview + create

The virtual network's address space, specified as one or more address prefixes in

IPv4 address space

10.0.0.0/16

☐ Add IPv6 address space ⓘ

The subnet's address range in CIDR notation (e.g. 192.168.1.0/24). It must be co

+ Add subnet

Remove subnet

Subnet name	Subnet address range
This virtual network doesn't have any subnets.	

✖ This virtual network doesn't have any subnets.

ⓘ A NAT gateway is recommended for outbound internet access from subnets. Edit th

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Add subnet

Subnet name *

subnet-1 ✓

Subnet address range * ⓘ

10.0.0.0/24 ✓

10.0.0.0 - 10.0.0.255 (251 + 5 Azure reserved addresses)

NAT GATEWAY

Simplify connectivity to the internet using a network address translation gateway. Outbound connectivity is possible without a load balancer or public IP addresses attached to your virtual machines. [Learn more](#)

NAT gateway

None ▾

SERVICE ENDPOINTS

Create service endpoint policies to allow traffic to specific azure resources from your virtual network over service endpoints. [Learn more](#)

Services ⓘ

0 selected ▾

Add

Cancel

4.1. Create WebSubnet

- Click Add Subnet
 - Subnet Name: WebSubnet
 - Subnet Address Range: 10.0.1.0/24
 - Click Add

4.2. Create AppSubnet

- Click Add Subnet
 - Subnet Name: AppSubnet
 - Subnet Address Range: 10.0.2.0/24

- Click Add

4.3. Create DbSubnet

- Click Add Subnet
 - Subnet Name: DbSubnet
 - Subnet Address Range: 10.0.3.0/24
 - Click Add

Step 5: Review and Create VNet

- Click Review + Create.
- Click Create.

Your VNet with three subnets is now ready.

Step 6: Verify Subnets

1. Go to your created VNet (ThreeTier-VNet).
2. Click on Subnets.
3. You should see all three subnets listed:

DEPLOYING VM IN EACH SUBNET

VM Creation Steps (Repeat for each VM)

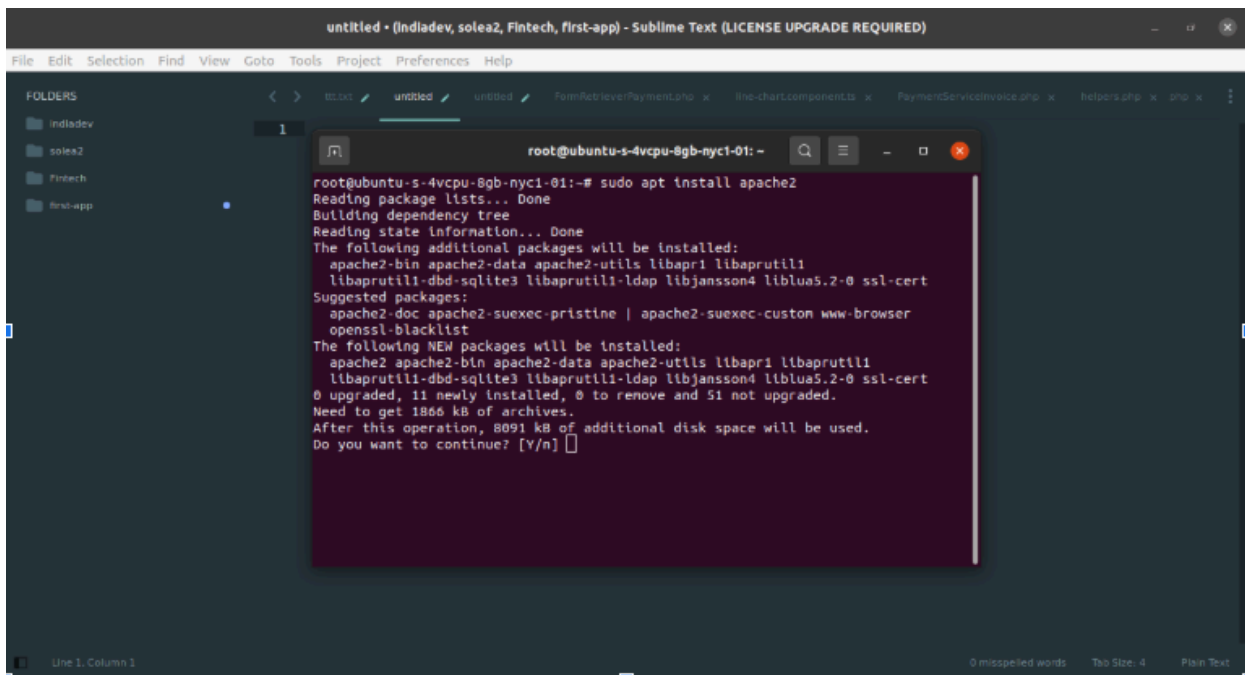
1. Search “Virtual Machines” → Create
2. Choose:

- Resource Group: ThreeTier-RG
- VM Name: <Your VM Name>
- Image: Ubuntu 20.04 LTS / Windows Server 2019
- Size: Standard_B2s (or per your need)
- Authentication: SSH for Linux, Password/RDP for Windows
- Public IP: Only for WebSubnet VMs
- NIC → Select correct subnet (WebSubnet, AppSubnet, DbSubnet)
- Assign NSG: No (subnet-level NSG is already configured)

3. Click Review + Create

Apache Setup on Linux VMs

SSH into Linux VMs:



The screenshot shows a Sublime Text editor window titled 'untitled - (Indiadev, solea2, Fintech, first-app) - Sublime Text (LICENSE UPGRADE REQUIRED)'. The editor has a sidebar on the left with a 'FOLDERS' pane showing a tree structure with folders 'indiadev', 'solea2', 'Fintech', and 'first-app'. The main editor area displays a terminal window with the following text:

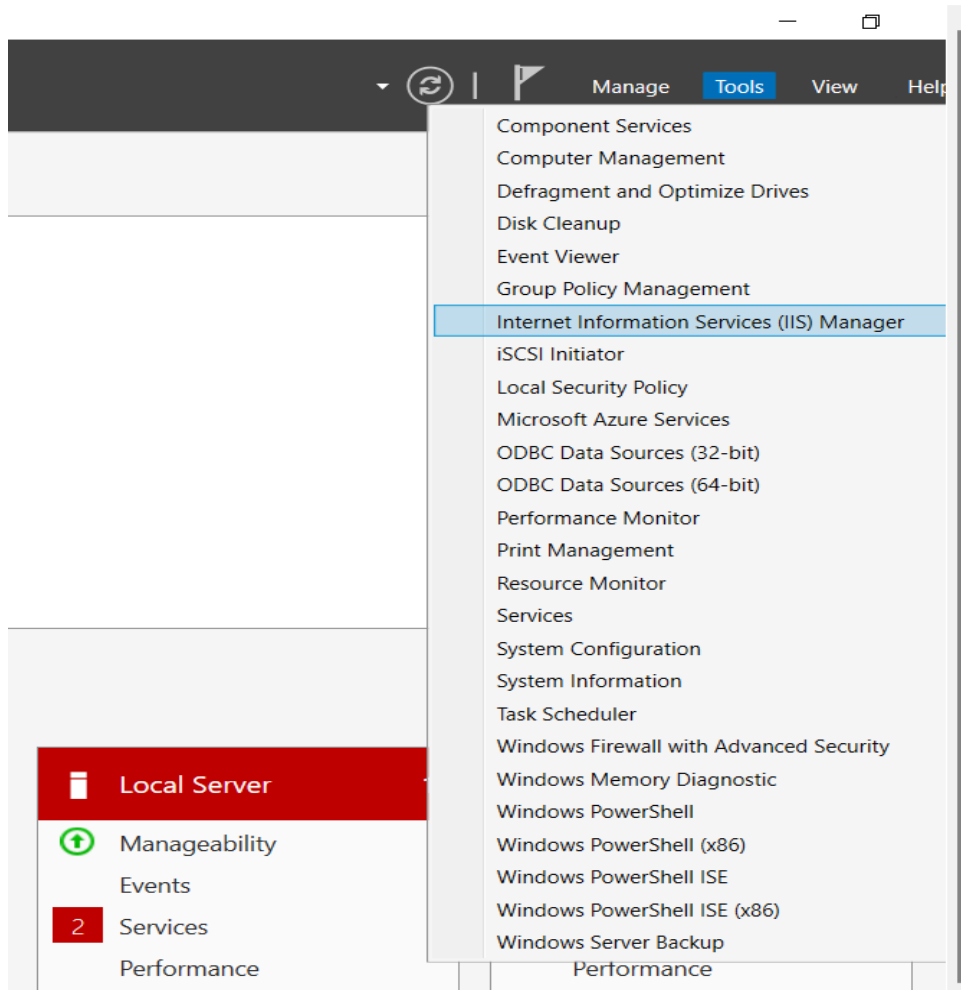
```
root@ubuntu-s-4vcpu-8gb-nyc1-01: ~
root@ubuntu-s-4vcpu-8gb-nyc1-01:~# sudo apt install apache2
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.2-0 ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
  openssl-blacklist
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap libjansson4 liblua5.2-0 ssl-cert
0 upgraded, 11 newly installed, 0 to remove and 51 not upgraded.
Need to get 1866 kB of archives.
After this operation, 8091 kB of additional disk space will be used.
Do you want to continue? [y/n]
```

Test: Open a browser → <http://<Public IP of Web-Linux-VM>>
(Default Apache page should load)

IIS Setup on Windows VMs

RDP into Windows VMs:

- Open PowerShell as Administrator



Test: Open browser → <http://localhost> or <http://<Public IP of Web-Windows-VM>>

Test Tier Communication

[Create a virtual machine](#) >

Networking

Basics Disks Networking Management Monitoring Advanced Tags Review

Configure network interfaces, virtual networks, public ip, network security groups, and load balancing.

[Learn more about basic virtual machine networking](#)

Network interface name ⓘ	web-linux-vm321
Virtual network ⓘ	ThreeTier-VNet
Subnet ⓘ	
Public IP ⓘ	New
NIC network security group ⓘ	None
Public inbound ports ⓘ	Would you like to enable public access ☐ None ☒ Enable Firewalls and security groups have different purposes. Learn more
Accelerated networking ⓘ	Disabled

[Review + create](#)

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[Next: Management >](#)

Next steps after configuration :

1. Complete the wizard for each VM:
 - Web tier: 1 Linux + 1 Windows
 - App tier: 1 Linux + 1 Windows
 - DB tier: 1 Linux + 1 Windows

2. Under Networking, ensure correct subnet selection, public IP assignment (if needed), and inbound ports.
 3. Click Review + create → Create.
 4. Repeat for all six VMs, adjusting names and subnet as above.
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